Business Problem Document

Airline Customer Insights: Surveys, Complaints and Passenger Data

Background

An international airline's Customer Care department collects passenger feedback from post-flight surveys (CSAT, NPS and Customer Effort Score) and handles customer complaint cases. Passenger volumes are recorded monthly by route and cabin class.

The data sits in a single Excel file that is maintained by hand. It contains duplicates, invalid scores, inconsistent spellings and text in numeric fields, so the KPIs reported from it are not reliable.

Note: this project uses a synthetic dataset created for practice. It does not contain real airline or passenger data.

Problem statement

Customer Care management does not have a single, trusted view of customer experience across routes, cabins and touchpoints. They cannot easily see:

- which routes and cabins have the lowest satisfaction and loyalty
- which issues drive dissatisfaction and complaints
- how efficiently complaints are resolved
- where to focus improvement efforts first

Key business questions

1. What are the overall CSAT %, NPS, CES, FCR % and complaints per 1,000 passengers?
2. Which routes and cabin classes perform worst on satisfaction and loyalty?
3. Which issue types have the lowest satisfaction and the most complaints?
4. Which routes have the most complaints relative to passenger volume?
5. Which complaint types are hardest to resolve at first contact, and how long do they take?
6. Are there repeat complainers who need service recovery?
7. Which data-quality problems exist, and how should they be fixed at the source?

Data

Table	Grain (one row =)	Key fields
Surveys	one survey response	survey_id, customer_id, survey_date, route, cabin_class, csat_score (1-5), nps_score (0-10), ces_score (1-7), issue_type
Complaints	one complaint case	case_id, customer_id, open_date, close_date, route, issue_type, contact_count, resolved
Flights	passengers per route, cabin and month	month, route, cabin_class, passengers

KPI definitions

KPI	Definition
CSAT %	Responses scoring 4-5 divided by all answered responses
NPS	% promoters (9-10) minus % detractors (0-6)
CES	Average effort score (1-7, lower is better)
FCR %	Cases solved in one contact and resolved, divided by all cases
Complaints per 1K pax	Complaints divided by passengers, times 1,000
Avg resolution days	Average days from open date to close date

Approach

1. **Python (Pandas):** clean the raw Excel file and record every fix in a data-quality log.
2. **PostgreSQL:** load the clean tables, build a star schema (Routes dimension, primary and foreign keys) and create KPI views.
3. **SQL:** answer the business questions.
4. **Power BI:** build an interactive dashboard (Overview, Complaints, Drivers).
5. **Presentation:** summarise findings and recommendations for management.

Expected outcome

A repeatable pipeline and dashboard that give Customer Care one trusted source for CX KPIs, clear priorities for improvement, and a list of data-quality fixes to agree with IT.